

30.4.15(E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
**(Autonomous College Affiliated to University of Mumbai)**

**End Semester Exam**  
**April - May 2015**

**Max. Marks: 100****Class: S.E****Name of the Course: Database Management System****Course Code: UCEC404****Duration: 3 hrs****Semester: IV****Branch: Computer****Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

| Question No. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Max. Marks |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q 1 (a)      | Discuss the main characteristics of database approach and how it differs from traditional file systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10         |
| Q 1 (b)      | Design a ER diagram for bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10         |
| Q2 (a)       | Explain the following relation algebra operations with example. <ol style="list-style-type: none"> <li>i. Left outer join</li> <li>ii. Generalized projection</li> <li>iii. Natural join</li> <li>iv. Select</li> <li>v. Rename</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10         |
| Q2 (b)       | <p>WORKS(Pname, Cname, Salary)<br/>           LIVES(Pname, Street, City)<br/>           LOCATED_IN(Cname, City)<br/>           MANAGER(Pname, Mgrname)</p> <p>Where Pname= Person name, Cname=Company name and Mgrname=Manager name</p> <p>Write SQL queries for the following:</p> <ol style="list-style-type: none"> <li>i. Find the names of the employees whose salaries are more than that of all of the 'Oracle' employees.</li> <li>ii. Find the names of companies that are located in every city where company 'Infosys' is located.</li> <li>iii. Give all employess of 'Wipro' company a 10% rise in salary.</li> <li>iv. List the names of the people who work for the company 'Wipro' along with city they live in.</li> </ol> | (3+3+2+2)  |
| Q3 (a)       | <p>Explain 2NF, 3NF and BCNF with suitable example.</p> <p align="center"><b>OR</b></p> <p>What is Multi-valued dependency? Explain 4NF with an example.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10         |
| Q3 (b)       | <p>Explain referential integrity and domain constraints with example.</p> <p align="center"><b>OR</b></p> <p>Discuss the mechanisms provided in SQL for defining authorization .</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10         |

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| Q4 (a) | <p>Define transaction. List ACID properties. Explain the usefulness of each.</p> <p style="text-align: center;"><b>OR</b></p> <p>Explain validation based concurrency control protocol.</p>                                         | 10 |
| Q4 (b) | <p>Explain heuristic approach of optimization with an example.</p> <p style="text-align: center;"><b>OR</b></p> <p>Explain log based recovery.</p>                                                                                  | 10 |
| Q5     | <p>Write short notes on <b>any 2</b></p> <ul style="list-style-type: none"> <li>i) Generalization and Specialization.</li> <li>ii) DBMS System architecture.</li> <li>iii) Database administrator and database designer.</li> </ul> | 20 |